

ARCHIT AGRAWAL

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EXPERIENCE

AI Software Engineer, EdPlus

Phoenix, AZ May 2026 – Present

- Architected and currently own a 12-agent analytics platform turning any structured dataset into charts and cited answers with zero per-dataset code; TypeScript, NestJS, AWS Bedrock, Step Functions.
- Built the Nuxt front end where a plain-English question returns an interactive chart with sort, filter, top-N and undo, on in-process DuckDB holding latency flat at 200x row volume.
- Designed a rules-based chart selector across 17 chart types grounded in visualization research; LLM suggestions ship only when they outscore the rules, so AI cannot worsen a chart.
- Quadrupled insight recall in automated reports, 13% to 52-59% on held-out datasets, by letting an agent pick the analyses while deterministic code computes every number.
- Caught a silent bug mislabeling 13% of rows before it reached institutional reports, reaching 100% join integrity with schema-constrained LLM output and citation checks.
- Benchmarked models against pipeline variants, finding model choice moved F1 33x more than architecture at \$5 to \$89 per 100k rows; redirected the roadmap to model selection.
- Reproduced the team's evaluation, traced a reported 0.637 accuracy to a hand-reviewed spreadsheet, and set a validated 0.523 baseline for future experiments.

Founding AI/ML Engineer, MyStage Music Inc.

Remote Jul 2025 – May 2026

- Re-architected a 14-Cloud-Function backend into one checkpointed LangGraph agent in Python on GCP Cloud Run, cutting production service dependencies 64%.
- Built a Playwright ingestion pipeline handling 70,000+ records a day from 650+ venues, with transactional locking against duplicate work and Gemini entity resolution raising dataset accuracy 25%.
- Delivered sub-50 ms search on FastAPI and Algolia with real-time Firestore sync, covered by 200+ pytest tests and end-to-end Logfire tracing.

Software Engineer, EdPlus

Phoenix, AZ Sep 2023 – May 2025

- Led end-to-end development of a multi-tenant RAG assistant, React front end to retrieval layer, used by 1,000+ faculty to author courses reaching 60,000+ students.
- Cut transcript analysis from 4 hours to 15 minutes, 16x, with a Neo4j knowledge graph queried by schema-validated LLM-generated Cypher.
- Enforced per-college data isolation at the retrieval layer so one tenant's material cannot surface in another's, making onboarding a config change, not a deployment.
- Built a Prompt Flow evaluation harness scoring groundedness, relevance and coherence, so every prompt change shipped on evidence rather than opinion.

Student Researcher, Arizona State University

Phoenix, AZ Aug 2024 – May 2025

- Built a PyTorch and Diffusers benchmark for text-to-video models scoring physical plausibility separately from visual quality, with seed, resolution and sampler held fixed.
- Showed visual fluency and physical plausibility correlate only weakly: the model ranked first on human preference ranked last on causality.

Data Research Aide, Knowledge Exchange for Resilience, Arizona State University

Phoenix, AZ Jun 2024 – Aug 2024

- Cut p95 retrieval latency 60% at 95% recall with a hybrid FAISS and Neo4j search layer.
- Built a FastAPI and pgvector recommendation service reaching 85% top-k precision for collaborator matching, fed by an ETL ingesting 10,000+ profiles a day, alongside 20+ REST APIs on .NET 8.

Software Engineer, Zeus Learning

Mumbai, India Jan 2022 – Jul 2023

- Built a Redis-backed desk-demand forecasting service used across 300+ Fortune 500 offices including Goldman Sachs and Merck, with reported desk occupancy up 30%.
- Cut deploy time 70% and production incidents 40% by containerizing the release pipeline and adding SonarQube quality gates scoped to new code.
- Migrated a C# and .NET monolith to Dockerized Kubernetes microservices incrementally behind the existing API, cutting infrastructure cost 20% and footprint 35%.

Product Intern, EAT.FIT

Bengaluru, India Sep 2021 – Dec 2021

- Cut driver-tracking server cost 45% with distance-based polling and client-side interpolation that kept map movement smooth.

EDUCATION

M.S., Computer Science, Arizona State University, GPA 4.0

Aug 2023 – May 2025

B.Tech., Information and Communication Technology, DA-IICT

Aug 2018 – May 2022